

ASE 420 Tools Installation

- Python
 - Python interpreter
 - Virtual environment and PIP

Python

- Install Python
 - [Download Python | Python.org](#)
- Always use a virtual environment (venv)
 - It isolates your working environment.
 - With PIP, you can install packages only for the venv.

- Learn how to use venv.
 - [Install packages in a virtual environment using pip and venv - Python Packaging User Guide](#)
- When you are using venv, the command line shows the directory of the venv: for example, (p3), at the prompt.

```
(p3) chos5@NKU023R7042 ase420>
```

- This is an example to start venv on Mac/Linux.

```
python -m venv ~/venv/p3  
source ~/venv/p3/bin/activate
```

- This is an example to start venv on Windows.

```
python -m venv C:\Users\YourUsername\venv\p3  
C:\Users\YourUsername\venv\p3\Scripts\activate.bat
```

Python Packages

- Under the virtual environment, use pip to install packages.
- Install pygame for the course project.
- Install jupyter for interactive Python application development.

```
pip install pygame  
pip install jupyter
```

When you have any issues

- Remember you are the problem solver.
- Develop your debugging skills as a professional software engineer.
- Start early to identify the issues as early as possible

Ask for help

- You don't have much time to lose, especially for the installation.
- Ask other team members who successfully installed Flutter.
- Use my office hours to visit and ask.

Discuss the issue with LLM

- Copy the error message to LLM.
- Use multiple LLMs when one LLM doesn't give you the correct answers.
- In the AI age, your time is more important than AI time.
- Learn to work with LLM , not following LLM's instructions; this is too dangerous.

Check if you have all the ASE Common tools

- Visual Studio Code
- Markdown/Marp
- Git/GitHub

Visual Studio Extension

- We use Visual Studio Code as the primary IDE.
 - Install Python extensions.
 - Install any extension for your application development.

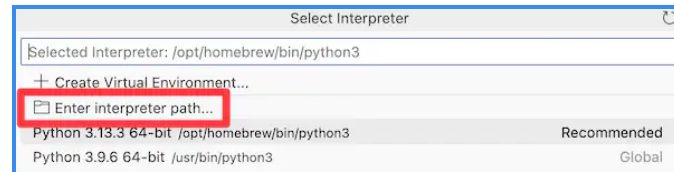
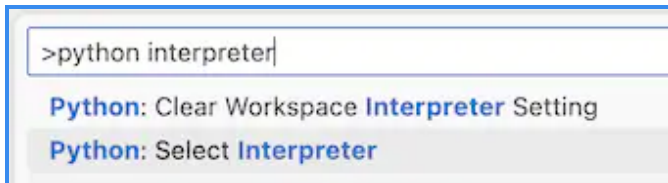
Run the Tetris programs

- git clone the ASE420 course from <https://github.com/nkuase/ase420>
- Using the command line, run the Python Tetris program in venv.

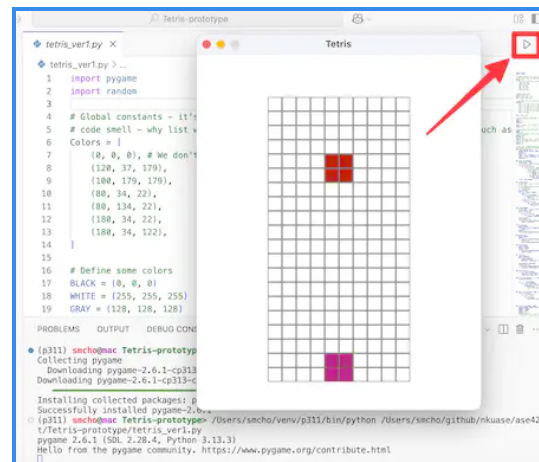
```
> cd project/Tetris-prototype  
> pip install pygame  
> python tetris_ver1.py
```

Run Tetris in VSC

- Install the Python extension in VS Code.
- Open the Tetris program in VSC.
 - Select Python interpreter and enter the Python of the venv.

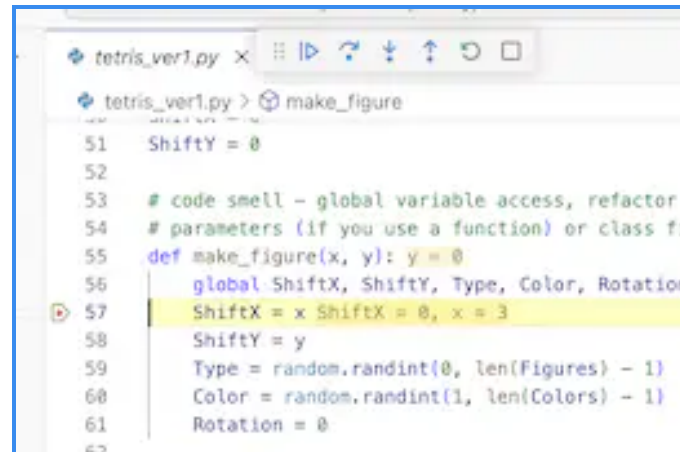
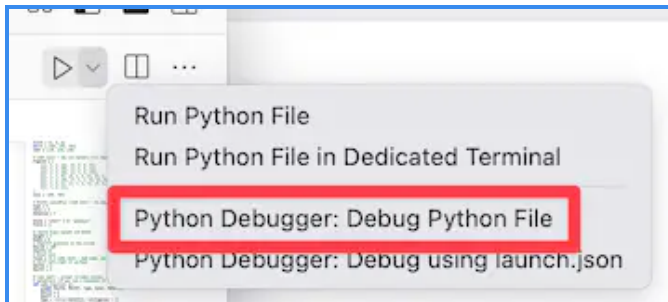


- Click the arrow to run the app.
- Install `pygame` using `pip` in your `venv` .



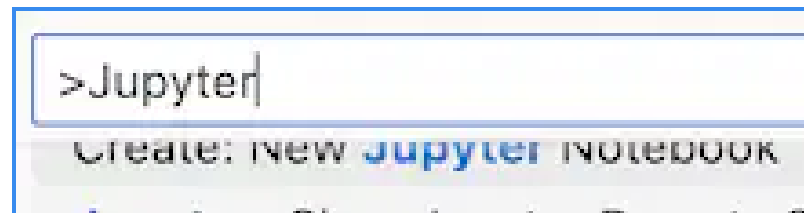
Debugging the App

- Use VSC: choose Python debugger and set breakpoints.



Use Jupyter

- Install Jupyter extension & set Python environment as your venv Python.



- Use Jupyter to learn Python and analyze the Tetris program.

